

Enhancing wood vinegar quality: Integrated pyrolysis process of cajuput (*Melaleuca leucadendron*) twigs biomass with gas-phase adsorption using zeolite adsorbents

Author links open overlay panelWahyu Anggo Rizal <sup>a b</sup>, [Satriyo Krido Wahono](#) <sup>b</sup>, Ahmad Tawfiequrrahman Yuliansyah <sup>a c</sup>

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Highlights

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Cajuput wastes from factories are utilized for wood [vinegar](#) production.

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Gas-phase adsorption during [pyrolysis](#) improves wood vinegar's quality.

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Zeolites decrease the levels of tar and PAHs.

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Wood [vinegar](#) comprises acids, [phenols](#), [furans](#), and pyran compounds.

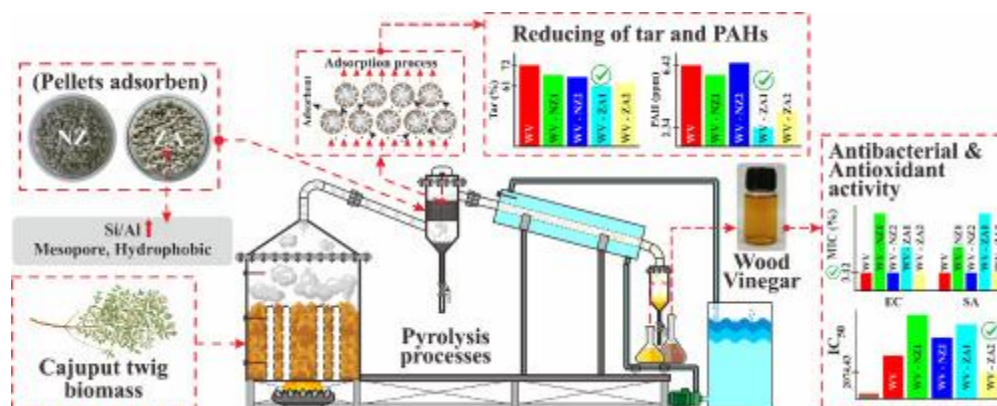
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Wood [vinegar](#) provides well ability as antibacterial and antioxidant properties.

Abstract

[Zeolite](#), being a porous material, can be used as an adsorbent. This study investigated the effect of [gas phase adsorption](#) using zeolite on wood [vinegar](#) (WV) characteristics from cajuput twig, which was pyrolyzed for 5 h at 500 °C. The WV composition of functional and toxic compounds was investigated using GC–MS. The antibacterial method involved [Escherichia coli](#) and [Staphylococcus aureus](#), whereas the [antioxidant activity](#) was determined by the [ABTS](#) method. Gas phase adsorption using zeolite can enhance the quality of WV, with the highest [MIC](#) indicating the two types of bacteria being 3.12 % and the highest [antioxidant activity](#) (IC<sub>50</sub>) being 2074.43 ppm. The functional compounds comprise acids, [phenols](#), [furans](#), and [pyrans](#). The reduction efficiency for tar was 14.61 %,

while toxic PAH compounds amounted to 79.45 %. Therefore, zeolite is recommended as a suitable adsorbent for WV production use in the [pyrolysis](#) system to reduce tar and PAH content.



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## Introduction

However, pyrolysis generates potentially harmful or carcinogenic substances, such as Polycyclic Aromatic Hydrocarbons (PAH), which are nonpolar and hydrophobic (Schlages et al., 2008). PAH is produced naturally from incomplete combustion, typically within the temperature range of 500 °C to 900 °C (Lingbeck et al., 2014; Šimko, 2005). PAH compounds can be reduced or eliminated by WV purification, allowing for adjustments in both color and taste intensity (Montazeri et al., 2013).

The common purification of WV distillation and adsorption techniques which utilizing natural zeolite, activated zeolite, silica gel, charcoal, and activated charcoal (Manni et al., 2007; Mansur et al., 2021; Maulina and Oktaviani, 2021). Typically, WV is purified through distillation and adsorption in its liquid phase. However, distillation requires more energy and time, and the liquid adsorption phase also takes longer. Zeolite finds extensive application in commercial processes, particularly as an adsorbent for removing PAH and other hazardous compounds (Hedayati et al., 2021; Mateen et al., 2015; Rad and Anbia, 2021; Solinska and Bajda, 2022).

In the previous research, modified zeolites were utilized as adsorbent in water media (liquid phase) to remove five types of PAH, achieving a removal rate of 84–94 % (Hedayati and Li, 2020). The adsorption using organo-zeolites of water vapor (gas phase) containing five types of PAH found that zeolite hydrophobicity greatly influences PAH compounds' adsorption ability (Lemić et al., 2007). Zeolites that contain tuff demonstrate a strong adsorption capacity for 10 types of PAH at liquid effluent (liquid phase) (Manni et al., 2007). Thus, zeolites have the potential to be adsorbent material in pyrolysis.

Natural zeolite exhibits varying structures, contingent upon the conditions of its natural formation. Therefore, utilizing zeolite as an adsorbent necessitates activation to enhance its physical attributes and eliminate impurities. This activation process can alter the Si/Al ratio and the type of cation present (Hernawan et al., 2015; Wahono et al., 2019a). Implementing direct gas-phase adsorption utilizing zeolite adsorbent materials integrated into a pyrolysis reactor could provide a more energy-efficient, time-saving for WV purification, ensuring the retention of its functional Properties while effectively eliminating hazardous compounds.

Based on these studies, this research aims to characterize and observe the physico-chemical properties of WV from cajuput twigs (CTB). The WV is produced by direct gas phase adsorption using zeolites in pyrolysis processes. It determines the impact on the yield of functional compounds and the reduction of harmful compounds in WV. WV's antibacterial properties and antioxidant studies are also conducted to determine potential WV application as food preservative material and achieve food-grade quality requirements.

## Material

The raw material for the pyrolysis experiment was biomass waste derived from the cajuput oil production process at the Sendang Mole factory in Yogyakarta, Indonesia (7°54'18.4"S 110°33'29.6"E). Zeolite was mined from Ponorogo, Indonesia (7.8667°S, 111.4666°E). Types of natural adsorbents (NZ) undertook activation (ZA) using hydrochloric acid (HCl p.a., 37–

38 %, Merck). Identification of PAH compounds was conducted using the PAH standards UST 122 (Merck) and dichloromethane (Merck).

#### Biomass characterization

Table 1 displays the results of biomass preparation and characterization. The CTB, which had a length of  $5 \pm 0.94$  cm, had an initial moisture content of 22.05 %. After drying, the moisture content decreased to 10.36 %. This low moisture content suits thermal conversion techniques and can affect biomass conversion (Mckendry, 2002). The ash content of CTB is 3.57 %, which is lower than *Melaleuca leucadendron* biomass (5.41 %) (Mansur et al., 2021) but higher than *Eucalyptus* sawdust (0.81 %) and

#### Conclusion

The pyrolysis of CTB produced WV with a main functional compound composition of acid and phenol. The addition of natural zeolite effectively increased acetic acid, while activated zeolite increased phenol in WV. The zeolite adsorbent material reduced tar by 14.61 % and PAH by 79.45 %. WV - ZA exhibited antibacterial properties with a MIC of 3.12 % against EC and SA bacteria. The highest  $IC_{50}$  value of 2074.43 ppm at WV - ZA2 indicated antioxidant properties. In general, natural zeolite was

#### CRedit authorship contribution statement

**Wahyu Anggo Rizal:** Writing – review & editing, Writing – original draft, Visualization, Resources, Formal analysis, Conceptualization. **Satriyo Krido Wahono:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Data curation. **Ahmad Tawfieurrahman Yuliansyah:** Writing – review & editing, Supervision, Methodology, Investigation, Data curation.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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